

Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 2

Summer Semester 2024-2025

DURATION: 25 MIN

FULL MARKS: 15

CSE 4803: Graph Theory

Figures in the right margin indicate full marks of questions with their corresponding COs and POs in parentheses.

1. A saturated hydrocarbon is a molecule of n Carbon atoms and k Hydrogen atoms such that all Carbon atoms are in 4 bonds and all Hydrogen atoms are in 1 bond, and no sequence of bonds forms a cycle, thus allowing the molecule to be represented by a tree. Prove that $k = 2n + 2$. 7
(CO2)
(PO2)
2. A k -regular graph is a graph where every vertex has a degree of k . Construct the smallest 3-regular graph with a vertex connectivity of 1, and show the construction process/logic. 8
(CO1)
(PO1)